

Program Pembangunan
Olimpiad Matematik (Siri 1)

Number Theory

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1 Divisibility

1.1 Theory

$$\begin{aligned}
 a \mid b &\iff a \text{ divides } b \quad (\text{i.e. } b \text{ is divisible by } a) \\
 &\iff b/a \in \mathbb{Z} \\
 &\iff b = ka \text{ for some } k \in \mathbb{Z} \\
 &\implies |a| \leq |b| \text{ or } b = 0
 \end{aligned}$$

1.2 Problems

1. Prove that $x \mid y$ and $y \mid z$ implies $x \mid z$.
2. Prove that $x \mid y$ and $x \mid z$ implies $x \mid ay + bz$.
3. If $n > 1$, prove that we can't have $n \mid 20n^2 + 25n + 1$.
4. If $a > b > 0$, prove that we can't have $a \mid 2025a + b$.
5. Find all positive integers n such that $n \mid 20n + 25$.
6. (AIME) What is the largest positive integer n for which $n^3 + 100$ is divisibly by $n + 10$?

2 Primes

2.1 Theory

p is prime $\iff$ its only divisors are 1 and p
 There are infinitely many primes.

2.2 Problems

1. Find all positive integers a such that $a^2 - 25$ is prime.
2. Find all positive integers a such that $n^2 - 10n + 24$ is prime.
3. (AMC) The sum of two prime numbers is 85. What is the product of these two prime numbers?
4. Find all positive integers n such that

$$3n - 4, 4n - 5 \text{ and } 5n - 3$$

are all prime.

3 Prime Factorisation

3.1 Theory

Fundamental Theorem of Arithmetic: Every integer $n > 1$ can be expressed uniquely into a product of primes, e.g.

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}.$$

- Divisors
- Number of Divisors
- Greatest Common Divisor
- Least Common Multiple
- *Euclid's Division Algorithm

3.2 Problems

1. Find the number of positive factors of 2025.
2. Find the number of positive factors of 2025 which are multiples of 3.
3. Find the number of positive factors of 2025 which are perfect squares.
4. Determine the smallest positive integer m such that $2m$ is a perfect square and $3m$ is a perfect cube.
5. (CJR) If m and n are positive integers such that $n^5 = 18000m$, find the smallest possible value of m .
6. What is the highest power of 3 dividing $7! + 8! + 9!$?
7. Prove that a positive integer n has an odd number of positive factors if and only if n is a perfect square.
8. Prove that $ab = \gcd(a, b) \times \text{lcm}(a, b)$.
9. For a prime p , prove that $\gcd(a, p) \in \{1, p\}$.

4 Modular Arithmetic

4.1 Theory

$$a \equiv b \pmod{n} \iff n \mid a - b$$

We can perform addition, subtraction and multiplication.

4.2 Problems

1. Find the remainder when 7^{2025} is divided by $2, 3, 4, \dots, 10$.
2. Prove divisibility rules.
3. Prove that $a - b \mid a^n - b^n$ for all positive integers n .
4. Prove that $a + b \mid a^n + b^n$ for all odd positive integers n .
5. Find all integers x such that $x \equiv 1 \pmod{2}$ and $x \equiv 2 \pmod{4}$.
6. Find all integers x such that $x \equiv 1 \pmod{2}$ and $x \equiv 2 \pmod{3}$.
7. If a and b are positive odd integers, prove that $a^2 + b^2$ is never a perfect square.
8. Prove that $2 + 5^n$ is never a perfect square for all $n \in \mathbb{Z}^+$.
9. Prove that the sequence $5, 12, 19, 26, 33, \dots$ contains no term of the form $2^n - 1$.
10. Prove that $x^2 + y^2 = 3^{2025}$ has no integer solutions.
11. Prove that $a^3 - a$ is always a multiple of 6.
12. Find all positive integers (a, b) such that $a^2 = b! + 2023$.
13. Find all positive integers (m, n) such that $1! + 2! + \dots + m! = n^2$.

5 Problems

1. (JBMO 2014 P1) Find all triples of primes (p, q, r) satisfying

$$3p^4 - 5q^4 - 4r^2 = 26.$$

2. (HMMT 2016 T1) Let a and b be integers (not necessarily positive). Prove that $a^3 + 5b^3 \neq 2016$.

3. (JBMO 2013 P1) Find all ordered pairs (a, b) of positive integers for which the numbers $\frac{a^3b-1}{a+1}$ and $\frac{b^3a+1}{b-1}$ are both positive integers.

4. (JBMO SL 2022 N1) Determine all pairs (k, n) of positive integers that satisfy

$$1! + 2! + \dots + k! = 1 + 2 + \dots + n.$$

5. (KJMO 1998 P1) Show that there exist no integer solutions (x, y, z) to the equation

$$x^3 + 2y^3 + 4z^3 = 9$$

6. (JBMO SL 2013 N1) Find all positive integers n for which

$$1^3 + 2^3 + \dots + 16^3 + 17^n$$

is a perfect square.

7. (JBMO SL 2011 N1) Solve in positive integers the equation

$$1005^x + 2011^y = 1006^z.$$